

Dual-Band, Bi-Directional Long-Haul Transmission in Reduced Outer Diameter Wideband HCF

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Abstract We demonstrate the first long-haul, bi-directional transmission using high-power bismuth- and erbium-doped fiber amplifiers in a novel recirculating-loop built on reduced-diameter HCF, transmitting 53.9+51.9 Tb/s (GMI) over 2024 km using O+C-bands, and 22.2+21.5 Tb/s (4500 km), 17.2+16.6 Tb/s (6750 km) using C-band only. ©2026 The Author(s)

Introduction

Anti-resonant (AR-) hollow-core fibers (HCFs) [1], have become attractive optical transmission media thanks to several advantages including low latency, low loss, low non-linearity, low chromatic dispersion and low back-scattering, enabling high launch powers, long span lengths and bi-directional transmission. These features have been successfully demonstrated for single-span systems [2-5], but repeated transmission over transoceanic distances remains relatively unexplored and, to date, has only been demonstrated for uni-directional systems.

In the C-band, transoceanic distances have been spanned with real-time [6] and offline [7] processing with throughputs of 10.3 Tb/s and 21.7 Tb/s respectively, over ~6600 km, and 24.8 Tb/s over 12,113 km [8]. In all cases, the achievable data-rate was limited by CO₂ absorption [9], which also further prevents expansion into the L-band where the absorption is 9-10 times stronger. This has limited dual-band C+L-band studies to 2500 km and comparatively low per-band data-rates [10]. Lower gas absorption in the O-band, together with the development of new amplifiers [11], has made this band the focus of spectrum expansion in HCF links, with long-haul transmission over 3000 km recently being demonstrated [12], enabled by bismuth-doped fiber amplifiers (BDFAs). However, these demonstrations used fibers with large outer diameters (ODs). In [6, 12] the fiber was 240 µm OD and the OD in [7] was not given, but thought to be similar or larger. Large ODs make it easier to achieve low losses, but larger area reduces the strand-count and ultimately limits cable capacity in space-constrained, long-haul scenarios.

Here, we address expanding spectrum and increasing spatial density of long-haul HCF transmission. We report the novel use of high power (HP) O-band BDFAs in combination with HP-

EDFAs to demonstrate combined bi-directional O- and C-band long-haul transmission using an ultra-wideband first window double-nested anti-resonant nodeless fiber (DNANF) HCF with a reduced OD of 190 µm. A 112.5 km fiber span (almost twice the 60 km span length used in [12]) was constructed from 5 spools with average cut-back measure losses of 0.16 dB/km at 1550 nm and 0.20 dB/km at 1310 nm. We transmitted C- and O-band signals in both directions of a recirculating loop set-up with combined 13 THz bandwidth, achieving a total data-rate of 53.9+51.9 Tb/s over 2024 km. We then focused on a C-band-only system, measuring the first bi-directional transmissions over transoceanic distances with throughputs of 22.2+21.5 Tb/s and 17.2+16.6 Tb/s at 4500 km and 6750 km, respectively. These results show that advanced HCF designs can enable long-haul dual-band, 2 amplifier systems with considerably more bandwidth than C+L-band systems in SMF or 2nd window HCF designs. Further, supporting bi-directional transmission combined with the 38% reduction in fiber area compared to previous HCF demonstrations [6, 12] means that this bandwidth may be delivered with greater spatial and spectral efficiency in next generation optical fiber systems.

Experimental set-up

The experimental set-up is shown in Fig. 1. The C- and O-bands used similar separate transceiver set-ups, shown in Fig. 1(a), each comprising a sliding 3-channel test band within a wideband WDM signal constructed from amplified spontaneous emission (ASE) noise shaped using optical processors (OPs). The test-band consisted of a 32 GBaud test-channel and 2 neighbouring channels (1.33 GHz guardband) originating from 10 kHz linewidth tunable lasers (TLs) for C-band channels and 200 kHz linewidth TLs for O-band signals. The combined test and dummy

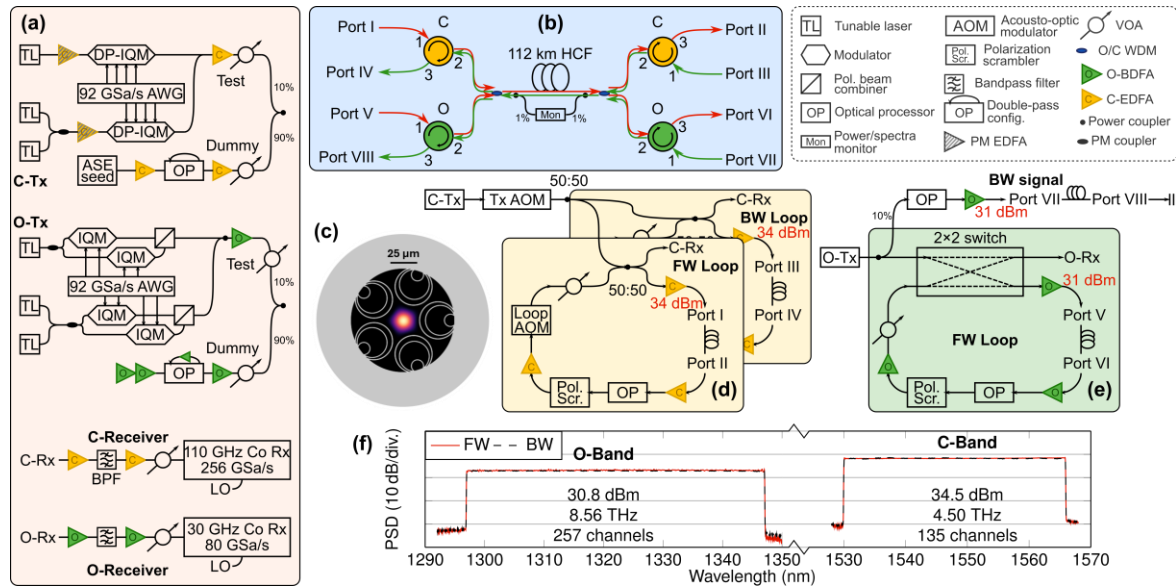


Fig. 1 Experimental set-up. (a) Transmitter and receiver layout, (b) Fiber span connections for dual-band Bi-Di transmission, (c) cross section of HCF, (d) C-band and (e) O-band loop configurations and (f) Bi-Di launched power.

channels of each band were split using a 3-dB coupler and used as the forward (FW) and backward (BW) signals for recirculating transmission.

The transmission fiber was a DNANF design with 5 sets of nested resonator tubes with the fiber profile shown in Fig. 1(c). The design is similar to the low-loss fiber reported in [6], but optimised to reduce the size of the microstructure and operating with a core size of 25 μm. The Bi-Di fiber link connections are shown in Fig. 1(b). Either side of the transmission fiber WDM couplers were used to separate/combine the O- and C-band signals with band-specific optical circulators then used to separate the forward and backward propagating signals. The total span loss including circulators, WDMs, HCF connectors, splices and fibers was 25 dB in the C-band and 30 dB in the O-band. The back reflected power was -59 dB at 1550 nm, and -60 dB at 1310 nm. The inter-modal interference (IMI) was measured to be lower than -60 dB in C-band and -55 dB in O-band. The CO₂ absorption at 195.35 THz peak was 0.05dB/km for the link. Fig. 2 shows the measured link loss including splices (~3 dB) and HCF-SMF connectors (~2 dB), averaged to per km loss, along with the chromatic dispersion, estimated from the DSP in each band. The C-band loop, shown in Fig. 1(d), used two fully recirculating loops, transmitting the signal FW and BW through the same span of HCF. The input signals shared the Tx load AOM, before FW and BW copies of the signal were produced using a 3 dB splitter and coupled into the respective loops. Each loop comprised HP booster of up to 34 dBm and low noise EDFAs at either side of the fiber, an OP for spectrum shaping and a polarization scrambler (PS). Zero-frequency-shifting AOMs were

used inside the loops to avoid unnatural smoothing of gas absorption lines. The O-band loop, shown in Fig. 1(e), transmitted signals through the same span of HCF as the C-band, and used a fast 2x2 electro-optic switch to route the signals into and out of the loop. In the absence of sufficient components to make two counter-propagating loops, a single loop contained HP-BDFA of up to 31 dBm [11], the HCF fiber span, BDFA, OP and PS for recirculation of the FW signal, while the counter propagating signal through the HCF used the continuous output from the transmitter, amplified and shaped to the equivalent power using an OP and HP-BDFA and terminated after transmission. Care was taken to ensure the transmitted signal spectrum was flattened to match the input spectrum at each recirculation. Figure 1(f) shows the optical spectra of the FW and BW signals launched into the fiber.

The signal in the receiver path of each direction was amplified and measured in each band separately. The CUT was filtered with an optical bandpass filter (BPF), attenuated with a variable optical attenuator (VOA), and detected using a coherent (Co) Rx employing a local oscillator (LO) of the same specifications as the transmitter lasers. 80 GSa/s and 256 GSa/s real-time oscilloscopes were used to acquire 10 traces for each

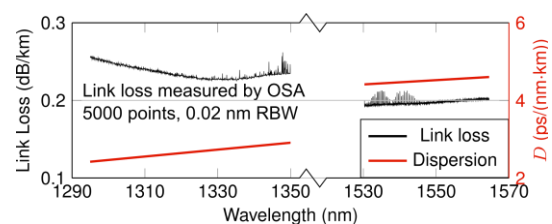


Fig. 2 Link loss and dispersion of the HCF.

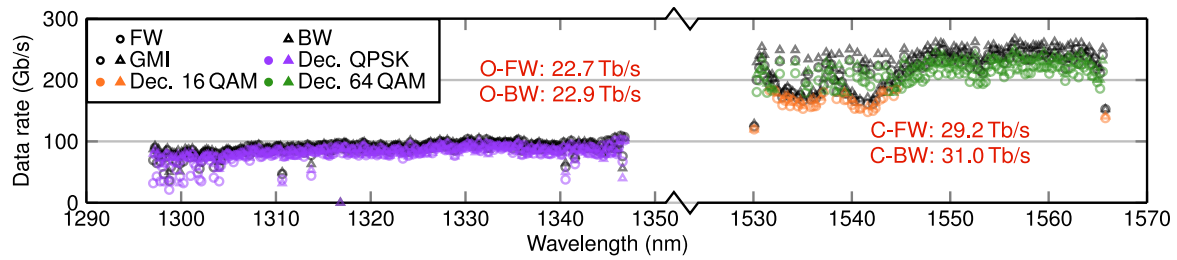


Fig. 3 Measured GMI-estimated and decoded data rates after 2024 km Bi-Di transmission.

signal channel in the O- and C-bands, respectively. Pilot-based DSP with 4% overhead and 101-tap equaliser for higher-order QAM, blind DSP with 41-tap decision-directed equaliser for QPSK signals, adaptive rate decoding, and code rate puncturing (with granularity of ~ 0.01) were applied offline [13]. An additional GLA compensation or spectrum flattening algorithm were used at the start of the DSP chain [14]. The generalized mutual information (GMI)-based and decoded data rates are reported for each channel by averaging the best 3 results out of 10 traces.

Results

Fig. 3 summarises the measured signal quality results after 2024 km Bi-Di transmission. The achievable information rate (AIR) estimated from GMI is shown for both bands and directions with the forward and backward transmission using circle and triangle markers, respectively. In the O-band, QPSK signals were used, while in the C-band, 16-QAM was used for the GLA-affected channels and 64-QAM elsewhere. In the O-band, most channels of the 8.56 THz bandwidth could support a net data-rate of approximately 100 Gb/s, but with some evidence of absorption lines at the longest wavelengths and deterioration of AIR at shorter wavelengths attributed to the increasing fiber loss. We note that, although the fiber loss in the O-band was only 0.20 dB/km, the additional components, connectors and splices contributed significantly to link loss and lead to relatively high ASE power that limited the overall O-band performance. We also note that many of the utilised components were not optimised for the O-band, so significant opportunity for improved performance exists.

In the C-band, a wider variation in AIR data-rates was observed, with the non-edge channels transmitting net rates from 150 Gb/s to 250 Gb/s. From the characteristic spectral shape, the

variation can be attributed to the impact of CO_2 absorption line in the shorter wavelength region. The total combined C- and O-band AIR was 51.9 Tb/s and 53.9 Tb/s (47.3 Tb/s and 49.7 Tb/s after decoding) in the FW and BW directions after 2024 km transmission with variation in the two directions potentially originated from differences in amplifier performance and losses in the circulators and WDM couplers at each end of the fiber.

C-band-only transmission results at longer distances of 4500 km and 6750 km are shown in Fig. 4. At 4500 km, the data-rate varied between 100 Gb/s and 200 Gb/s with lower AIR channels located around the two CO_2 absorption peaks. At this distance the total data-rate was 21.5 Tb/s and 22.2 Tb/s (19.8 Tb/s and 20.6 Tb/s after decoding) in the two directions. At the longest distance of 6750 km, a similar variation in the channel AIRs was observed, but with the channels most affected by gas absorption becoming nearly unrecoverable. The total FW and BW AIRs were 16.6 Tb/s and 17.2 Tb/s (15.1 Tb/s and 15.6 Tb/s after decoding), respectively.

Conclusions

We have demonstrated the first long-distance bi-directional optical fiber transmission utilizing a 112.5 km span of novel wideband anti-resonant hollow-core fiber with a reduced 190 μm outer diameter and 38% smaller fiber area than previous HCF demonstrations. We measured combined C+O-band GMI data-rates of 53.9 Tb/s + 51.9 Tb/s in FW and BW directions at a distance of 2024 km and C-band data-rates of 22.2 + 21.5 Tb/s at 4500 km and 17.2 and 16.6 Tb/s at 6750 km. These results show that advanced AR-HCFs supporting dual-band, bi-directional transmission can bring the low-latency transmission into the long-haul regime and support high-data-rates in space-constrained regimes.

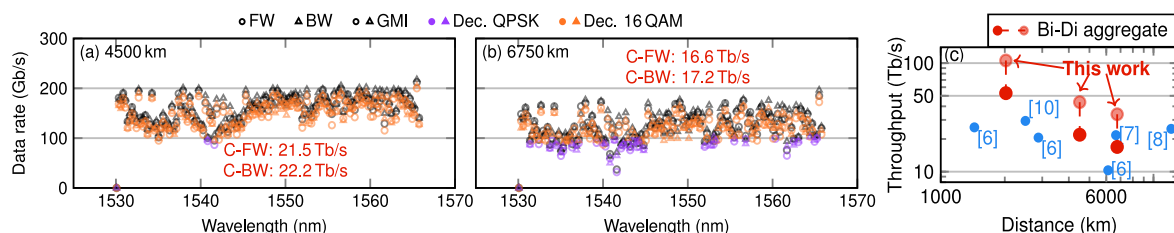


Fig. 4 (a) 4500 km and (b) 6750 km C-band Bi-Di transmission results. (c) State-of-the-art long-haul HCF demonstrations.

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